2.5a Multiply out and simplify to obtain SOP

$$(A+B)(C+B)(D'+B)(ACD'+E)$$

Note: this is in POS form
So, to convert to SOP form, let us use
2nd DL $(X+Y)(X+Z) = X+YZ$

$$F = \underbrace{(A+B)(C+B)}(D'+B)(ACD'+E)$$ 2nd DL

$$= \underbrace{(AC+B)(D'+B)}(ACD'+E)$$ 2nd DL

$$= (A\underbrace{C \cdot C}D'+B)(ACD'+E)$$ Idempotent law

$$= (ACD'+B)(ACD'+E)$$

Answer

$\boxed{F = ACD' + BE}$ which is in SOP form

2.6c Factor to obtain POS

$$G = A'BC + EF + DEF'$$

This is in SOP form
So, to convert to POS form, let us use
1st DL : $X \cdot Y + X \cdot Z = X \cdot (Y+Z)$
2nd DL : $X + YZ = (X+Y)(X+Z)$

$$G = A'BC + EF + DEF'$$ 1st DL

$$= A'BC + E(F + DF')$$ 2nd DL

$$= A'BC + E(F+D)(F+F')$$ Idempotent law

$= A'BC + E(F+D)\cdot 1$ 2nd DL

$= (A'BC+E)(A'BC+F+D)$ 2nd DL

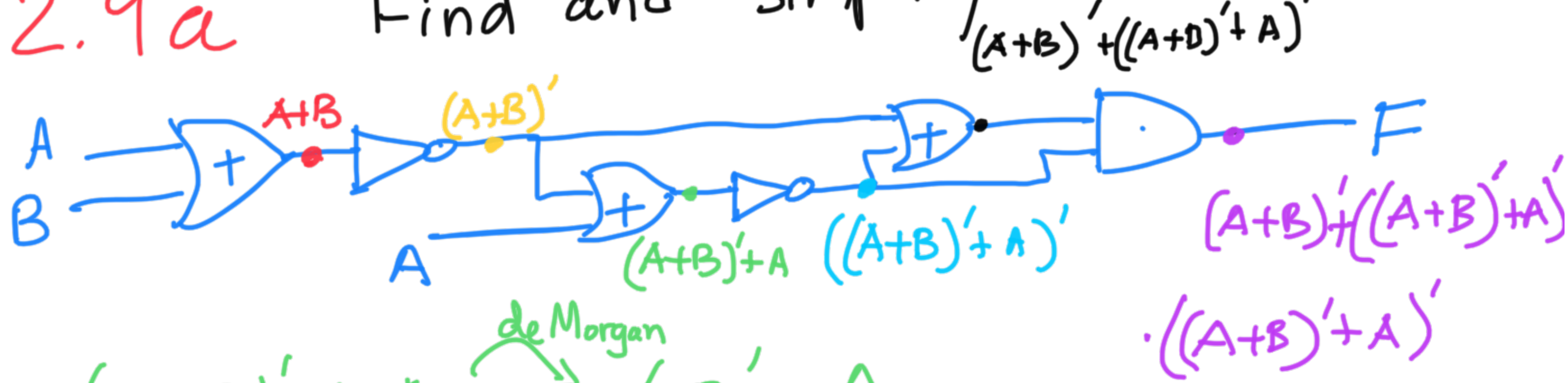
$= (A'+E)(BC+E)(A'+F+D)(BC+F+D)$ 2nd DL

Answer

$$\boxed{G = (A'+E)(B+E)(C+E)(A'+F+D)(B+F+D)\cdot(C+F+D)}$$

which is in POS form

2.9a Find and simplify



$(A+B)'+A \overset{\text{deMorgan}}{=} A'\cdot B'+A$

$= A+B'$ Elimination Thm.

$((A+B)'+A)' = (A+B')'$

$(A+B)'+((A+B)'+A)' = (A+B)'+(A+B')'$

$= A'B'+A'B$ deMorgan

$= A'(B'+B)$ 1st DL

$= A'$ Idempotent

$F = A'\cdot(A+B')'$

$= A'\cdot A'B$ deMorgan

$= A'B$ Idempotent

$$\boxed{F = A'B}$$